

MATH 130: 9/15 WORKSHEET

A key observation about logic is that what is relevant isn't the *content* of statements but their *form*.

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|---|---------------------------------------|--|
| • Either Seth Moulton won the senate primary or else Ed Markey won. | • Either 3 is even or else it is odd. | • Either it is raining or else it's snowing. |
| • Moulton didn't win. | • 3 is not even. | • It's not raining. |
| • $\therefore$ Markey won. | • $\therefore$ 3 is odd. | • $\therefore$ It is snowing. |

All of these are valid. The meaning doesn't determine that, only the form.

- Either A or B .
- Not A .
- $\therefore B$.

It is convenient to have a formal language to allow us to analyze form without distraction or ambiguity. In jargon, we want a language for *propositional logic*.

Propositional variables.

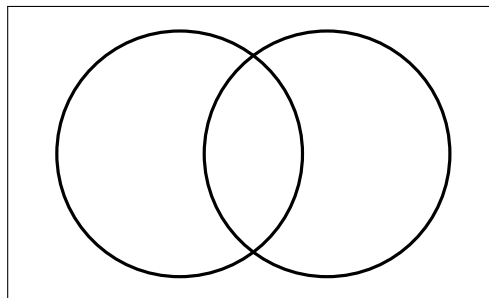
We use capital letters $A, B, P, Q, \dots$ for *propositional variables*. These are variables that stand for an *atomic proposition*—something that can't be broken down any further.

Sometimes we work abstractly, with no meaning in mind for the propositional variables. If you do have a meaning in mind, you should give a *translation key* which gives their meaning.

There are multiple different ways to combine atomic propositions to build up more complex propositions. We will take five of them as basic.

Conjunction (and).

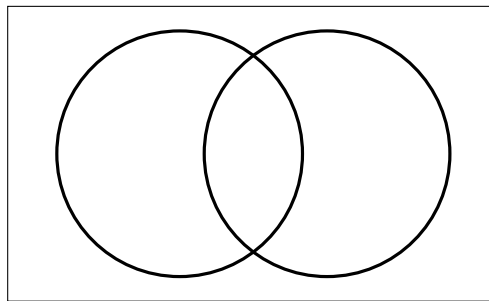
$A \wedge B$ means "both A and B ".



Disjunction (or).

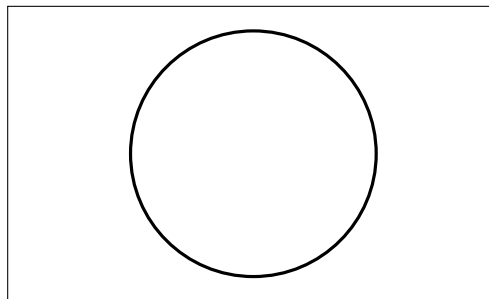
$A \vee B$ means “ A or B , possibly both”.

- This is an *inclusive* or, we allow both to be true.
- An *exclusive* or, where you don’t allow both at once, could also be used. But it’s a different concept and would need a different symbol.

**Negation (not).**

$\neg A$ means “not A ” or “ A is not the case”.

- This is the only *unary* connective, meaning it has a single input.
- The others are *binary* connectives, meaning they have two inputs.

**Conditional (if-then).**

$A \Rightarrow B$ means “if A then B ”.

There are many ways to say this.

- A implies B .
- B only if A .
- A is a sufficient condition for B .
- B is a necessary condition for A .

**Biconditional (if-and-only-if).**

$A \Leftrightarrow B$ means “ A if and only if B ”, which is the same as “if A then B and also if B then A ”.

- A common abbreviation for “if and only if” is “iff”.



Parentheses are used to express *priority* for complicated propositions with multiple connectives.